CSEP 590tv: Quantum Computing

Dave Bacon

June 22, 2005





Today's Menu

Administrivia

What is Quantum Computing?

Quantum Theory 101

Quantum Circuits

Linear Algebra

Administrivia

Le Syllabus

Course website: http://www.cs.washington.edu/csep590
[power point, homework assignments, solutions]

Mailing list: https://mailman.cs.washington.edu/csenetid/auth/mailman/listinfo/csep590

Lecture: 6:30-9:20 in EE 01 045

Office Hours:

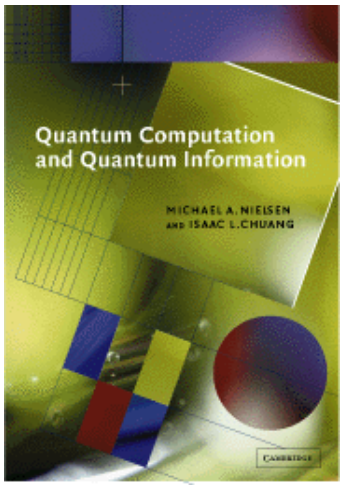
Dave Bacon, Tuesday 5-6pm in 460 CSE
Ioannis Giotis, Wednesday 5:30-6:30pm in TBA

Administrivia

Textbook:

"Quantum Computation and Quantum Information"
by Michael Nielsen and Isaac Chuang







Supplementary Material:

John Preskill's lecture notes
http://www.theory.caltech.edu/people/preskill/ph229/

David Mermin's lecture notes
http://people.ccmr.cornell.edu/~mermin/qcomp/CS483.html

Administrivia

Homework: due in class the week after handed out

1. Extra day if you email me
2. One homework, one full week extension, email me
3. Major obstacles, email me
4. Collaboration fine, but must put significant effort on your own first and write-up must be “in your words.”

Final Take Home Exam

Making the Grade: GRADES!!!!

70% Homework, 30% Final

Administrivia

Quick survey

Linear Algebra: all Do You Remember It: 50%

Quantum Theory: ¼ remember: 0

Computational Complexity: ¼

Background:

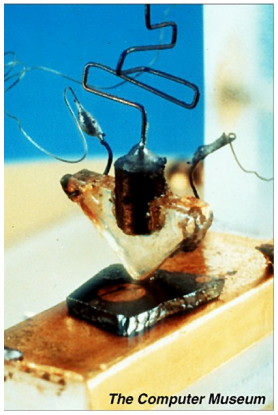
Computer Science:2/3
Computer Engineering: 4 peebs
Electrical Engineering: 1
Physics: 3
Other: 0

In the Beginning…



Alan Turing

1936- “On computable numbers, with an application to the Entscheidungsproblem”



1947- First transistor



1958- First integrated circuit

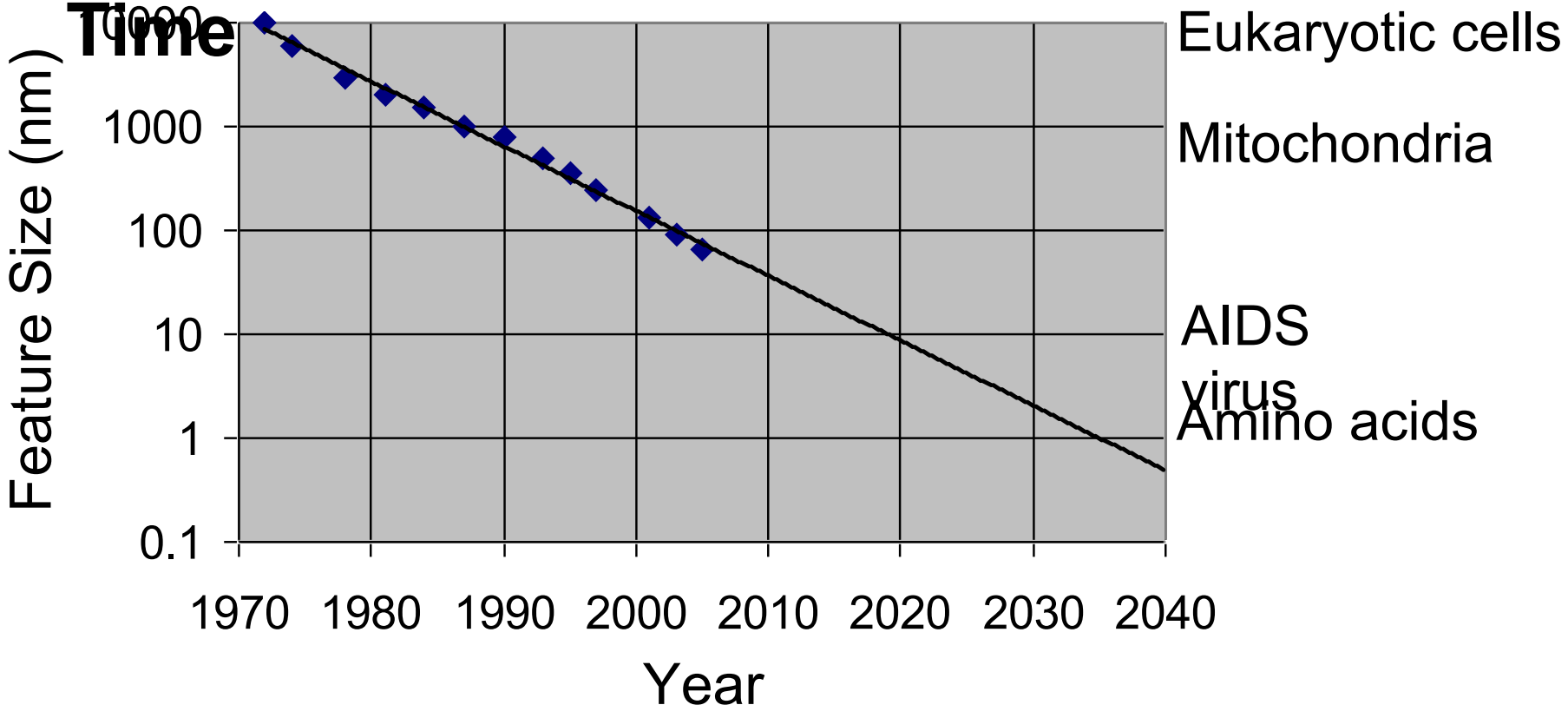


1975- Altair 8800



2004
GHz machines that weight ~ 1 pound

Moore's Law


Computer Chip Feature Size versus Time
Feature Size (nm)
10000
1000
100
10
1
0.1
1970
1980
1990
2000
2010
2020
2030
2040
Year
Eukaryotic cells
Mitochondria
AIDS virus
Amino acids

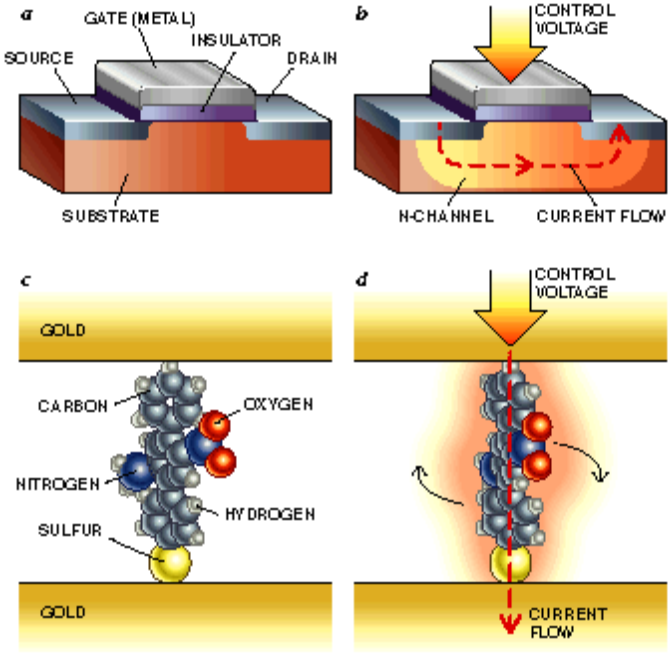
This Is the End?

1. Ride the wave to atomic size computers?

2. How do machines of atomic size operate?

Argument by Unproven Technology

1. Ride the wave to atomic size computers?



molecular transistors

Pic: http://www.mtmi.vu.lt/pfk/funkc_dariniai/nanostructures/molec_computer.htm

This Is the End?

2. How do machines of atomic size operate?

“Quantum Laws”

“Classical Laws”



“Quantum Computers?”

This Is the End?

2. How do machines of atomic size operate?



Richard Feynman

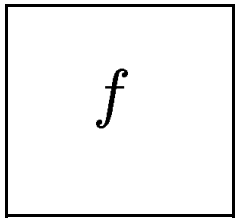


David Deutsch



Paul Benioff

Query Complexity



$$f: \quad \{0,1\}^n \quad \rightarrow \quad \mathcal{S}$$

n bit strings — set

$$P(f): f \rightarrow \mathcal{T}$$

set of properties

How many times do we need to query f in order to determine $P(f)$?

Example $f: \{0,1\}^n \rightarrow \{0,1\}$:

$$P(f) = \begin{cases} 0 & \text{if } f(x) = 0, \forall x \\ 1 & \text{if otherwise} \end{cases}$$

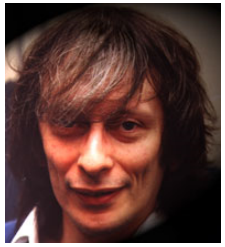
Promise problem: restricted set of functions domain of P not all f

The Work of Crazies



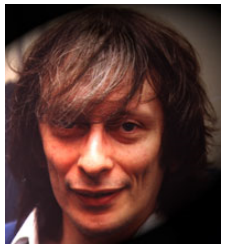
Richard Feynma

"Can Quantum Systems be Probabilistically Simulated by a Classical Computer?"



David Deutsch

1985: two classical queries
one quantum query
(but sometimes fails)



David Deutsch



Richard Jozsa

1992: $O(2^n)$ classical queries

2 quantum queries

$O(\log(p^{-1}))$classical queries to solve with probability of failure p

Crazies…Still Working



Umesh Vazirani



Ethan Bernstein

1993 : superpolynomially more classical than quantum queries



Dan Simon

1994 : exponentially more classical than quantum queries

The Factoring Firestorm



Peter Shor 1994

1881988129206079633386972394616504 3
9807163563379417382700763356422988 8
5971523466548531906060650474304531 7
3880113033967161996923212057340318 7
9550656962213051687593076502570 59

=

3980750864240649373971
2550055038649119906436
2342526708406385189575
9463889572617685833 17

×

4727721461074353025362
2307197304822463291469
5302097116459852171130
5207112563635903975 27

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's quantum algorithm takes time $O(n^2 \log n)$

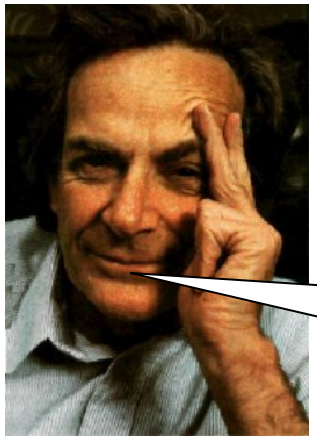
An efficient algorithm for factoring **breaks** the RSA public key cryptosystem

This Course

1. Quantum theory the easy way
2. Quantum computers
3. Quantum algorithms (Shor, Grover, Adiabatic, Simulation)
4. Quantum entanglement
5. Physical implementations of a quantum computer
6. Quantum error correction
7. Quantum cryptography

Quantum Theory

Slander



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Anyone who is not shocked by quantum theory has not understood it.



Niels Bohr
Nobel Prize 1922

Quantum Th



Electromagnetis
m

Strong force

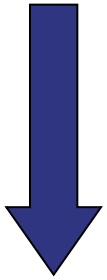
Weak force

Gravity (?)

“Quantum theory is the machine language of the universe”

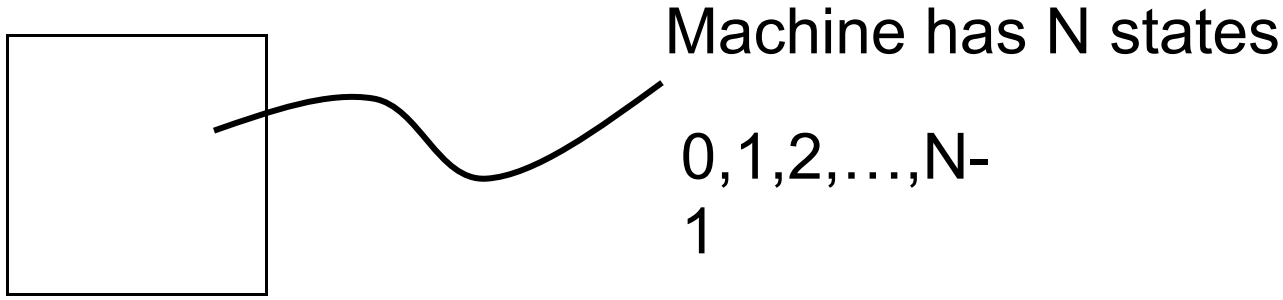
Our Path

Probabilistic information processing device



Quantum information processing device

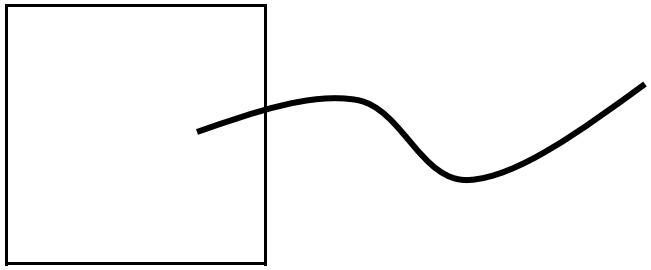
Probabilistic Information Processing Device



Rule 1 (State Description)

A probabilistic information processing machine is a machine with a state labeled from a finite alphabet of size N. Our description of the state of this system is a N dimensional real vector with positive components which sum to unity.

Rule 1



Machine has N states

0,1,2,…,N-1

N dimensional real vector

$$p \in \mathbb{R}^N \qquad p = \begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_{N-1} \end{bmatrix}$$

positive elements

$$p_i \geq 0$$

which sum to unity

$$\sum_{i=0}^{N-1} p_i = 1$$

probability vector

Example: 3 state device

$$p = \begin{bmatrix} 0.3 \\ 0.7 \\ 0 \end{bmatrix}$$

30 % state 0

70 % state 1

0 % state 2

Probabilistic Information Processing Device

Rule 1 (State Description) N states, probability vector

Rule 2 (Evolution)

The evolution in time of our description of the device is specified by an N x N stochastic matrix A, such that if the description of the state before the evolution is given by the probability vector p then the description of the system after this evolution is given by q=Ap.

Rule 2

Evolution:

If we are in state 0, then with probability Aj,0 switch to state j

If we are in state 1, then with probability Aj,1 switch to state j

⋮

If we are in state N, then with probability Aj,N switch to state j

N2 numbers

probability to be in state j after evolution $q_j = \sum_{i=0}^{N-1} A_{j,i} p_i$

$$\begin{bmatrix} q_0 \\ \vdots \\ q_{N-1} \end{bmatrix} = \begin{bmatrix} A_{0,0} & \cdots & A_{0,N-1} \\ \vdots & & \vdots \\ A_{N-1,0} & \cdots & A_{N-1,N-1} \end{bmatrix} \begin{bmatrix} p_0 \\ \vdots \\ p_{N-1} \end{bmatrix}$$

$$q = Ap$$

Rule 2

$$q_j = \sum_{i=0}^{N-1} A_{j,i} p_i$$

these are probabilities

$$A_{j,i} \geq 0$$

$$\sum_{j=0}^{N-1} A_{j,i} = 1$$

stochastic matrix

If in state 0 switch to state 0 with probability 0.4

If in state 0 switch to state 1 with probability 0.6

If in state 1 always stay in state 1

$$A = \begin{bmatrix} 0.4 & 0 \\ 0.6 & 1 \end{bmatrix} \quad p = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} \qquad \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} \rightarrow \begin{bmatrix} 0.16 \\ 0.82 \end{bmatrix}$$

$$q = Ap = \begin{bmatrix} 0.4 & 0 \\ 0.6 & 1 \end{bmatrix} \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 0.16 \\ 0.84 \end{bmatrix}$$

Probabilistic Information Processing Device

Rule 1 (State Description) N states, probability vector

Rule 2 (Evolution) N x N stochastic matrix

Rule 3 (Measurement)

A measurement with k outcomes is described by k N dimensional real vectors with positive components. If we sum over all of these k vectors then we obtain the all 1's vector. If our description of the system before the measurement is p, then the probability of getting the outcome corresponding to vector m is the dot product of these vectors. Our description of the state after this measurement is given by the point wise product of the outcome vector with p, divided by the probability of obtaining the outcome.

Rule 3

Simple measurement: If we simply look at our device, then we see the states with the probabilities given by the probability vector.

$$p = \begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_{N-1} \end{bmatrix}$$

More complicated measurements:
measurements which don't fully distinguish states

Example: if state is 0 or 1, outcome is 0
if state is 3 or 4, outcome is 1

measurements which assign probabilities of outcomes for a given state measurement

Example: if state is 0, 40% of the time outcome is 0
and 60% of the time outcome is 1
if state is 1, outcome is always 1

Rule 2

Measurement ➡ k vectors

$$m_\alpha \in \mathbb{R}^N$$

$$\alpha = 0, 1, \ldots, k-1$$

α: measurement outcomes

Probability of outcome α

$$\boxed{Pr(\alpha) = p \cdot m_\alpha} = \sum_{i=0}^{N-1} p_i (m_\alpha)_i$$

Require that these are probabilities

$$\sum_{\alpha=0}^{k} Pr(\alpha) = \sum_{\alpha=0}^{k} p \cdot m_\alpha = \sum_{i=0}^{N-1} p_i (m_\alpha)_i = 1$$

$$\boxed{(m_\alpha)_i \geq 0}$$

$$\boxed{\sum_{\alpha=1}^{k-1} (m_\alpha)_i = 1}$$

Rule 3 Update Rule

What is the probability vector after a measurement? $p \longrightarrow q$

Bayes' Rule:

$$Pr(A|B) = \frac{Pr(B|A)Pr(A)}{Pr(B)}$$

B := outcome α A := being in state j

$$\boxed{q_j = \frac{(m_\alpha)_j p_j}{(m_\alpha \cdot p)}}$$

$(m_\alpha)_j$ are conditional probabilities of being in state j given outcome α

Valid probabilities:

$$\sum_{j=0}^{N-1} q_j = \sum_{j=0}^{N-1} \frac{(m_\alpha)_j p_j}{(m_\alpha \cdot p)} = 1$$

Rule 3 In Action

Two state machine with probability vector: $p = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}$

Three outcome measurement *(k=3)*

$$m_0 = \begin{bmatrix} 0.3 \\ 0 \end{bmatrix} \quad m_1 = \begin{bmatrix} 0.7 \\ 0.2 \end{bmatrix} \quad m_2 = \begin{bmatrix} 0 \\ 0.8 \end{bmatrix}$$

Probability of these three outcomes:

$$Pr[0] = 0.3 \times 0.4 + 0 \times 0.6 = 0.12$$
$$Pr[1] = 0.4$$
$$Pr[2] = 0.48$$

Outcome 0:
$$q = \begin{bmatrix} \frac{0.3 \times 0.4}{0.12} \\ \frac{0 \times 0.6}{0.12} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Outcome 1
$$q = \begin{bmatrix} \frac{0.7 \times 0.4}{0.4} \\ \frac{0.2 \times 0.6}{0.4} \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}$$

Outcome 2:
$$q = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Probabilistic Information Processing Device

Rule 1 (State Description) N states, probability vector

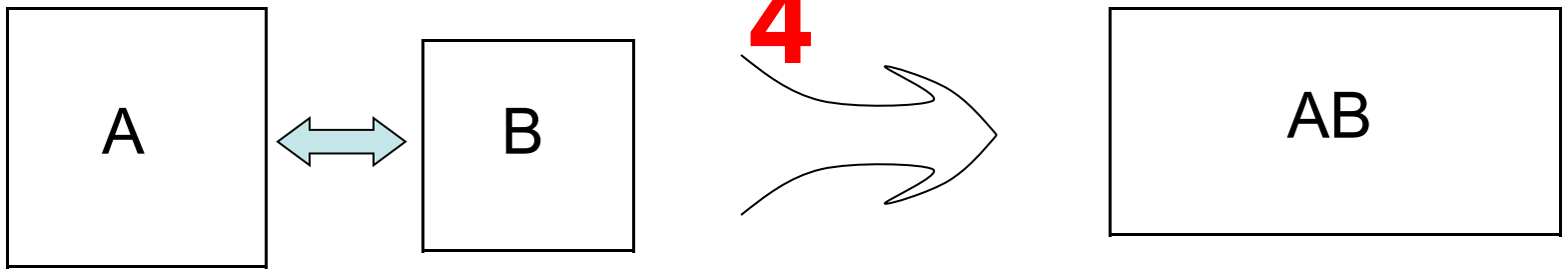
Rule 2 (Evolution) N x N stochastic matrix

Rule 3 (Measurement) k conditional probability vectors

Rule 4 (Composite Systems)

Two devices can be combined to form a bigger device. If these devices have N and M states, respectively, then the composite system has NM states. The probability vector for this new machine is a real NM dimensional probability vector from $\mathbb{R}^N \times \mathbb{R}^M$.

Rule 4



N States

0
1
⋮
N

M States

0
1
⋮
M

NM States

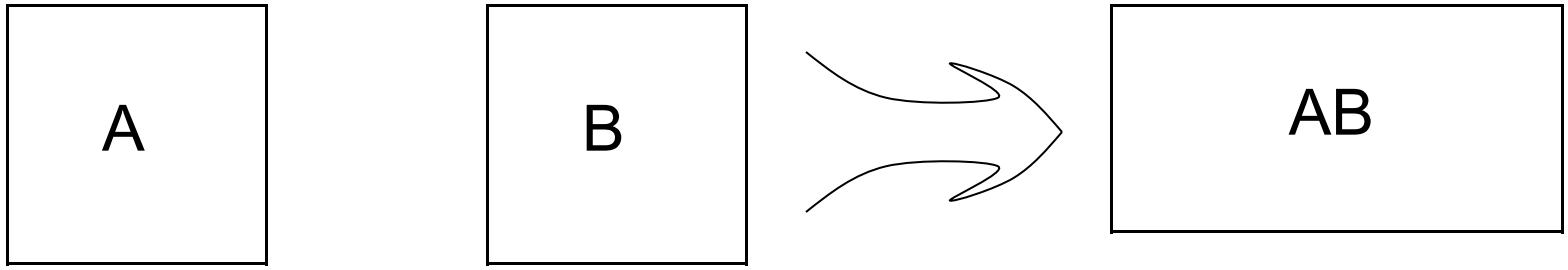
0,0	1,0	••	N,0
0,1	1,1		N,1
⋮	⋮		⋮
0,M	1,M	••	N,M

Probability vector in $\mathbb{R}^N \times \mathbb{R}^M$

$$\mathbb{R}^N \times \mathbb{R}^M = \mathbb{R}^{NM}$$

$$\mathbb{R}^N \times \mathbb{R}^M \neq \mathbb{R}^{N+M}$$

Rule 4 In Action



$$p = \begin{bmatrix} p_1 \\ 1-p_1 \end{bmatrix} \quad q = \begin{bmatrix} q_1 \\ 1-q_1 \end{bmatrix} \qquad r$$

$$r = p \otimes q = \begin{bmatrix} p_1 \\ 1-p_1 \end{bmatrix} \otimes \begin{bmatrix} q_1 \\ 1-q_1 \end{bmatrix} = \begin{bmatrix} p_1 q_1 \\ p_1(1-q_1) \\ (1-p_1)q_1 \\ (1-p_1)(1-q_1) \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$

contrast with $r = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$.

Probabilistic Information Processing Device

Rule 1 (State Description) N states, probability vector

Rule 2 (Evolution) N x N stochastic matrix

Rule 3 (Measurement) k conditional probability vectors

Rule 4 (Composite Systems) tensor product

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

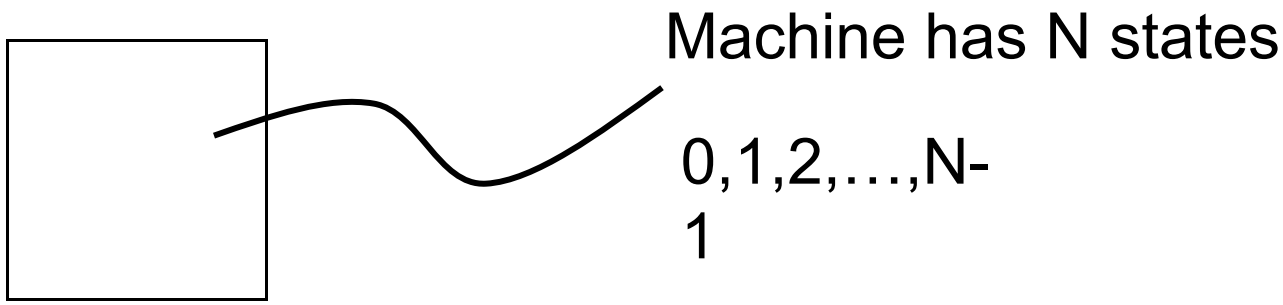
Rule 2 (Evolution) N x N unitary matrix

Rule 3 (Measurement) k measurement operators

Rule 4 (Composite Systems) tensor product

Quantum Rule 1

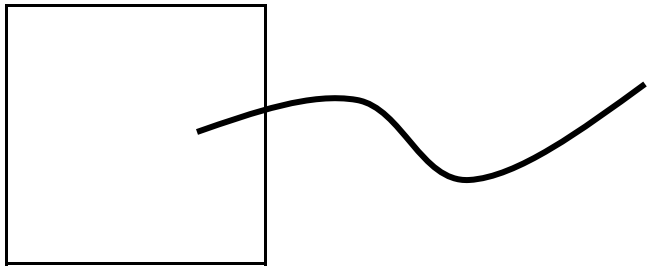
Rule 1 (State Description)



Rule 1 (State Description)

A quantum information processing machine is a machine with a state labeled from a finite alphabet of size N. Our description of the state of this system is a N dimensional **complex unit** vector

Quantum Rule 1



Machine has N states

0,1,2,…,N-1

N dimensional complex vector (vector of amplitudes)

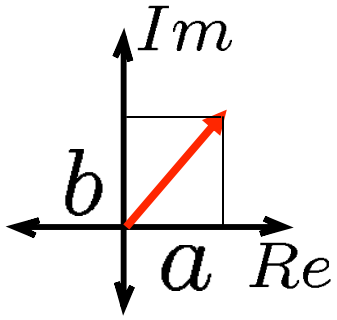
$$v \in \mathbb{C}^N \qquad v = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \qquad v_i \in \mathbb{C}$$

Complex numbers:

$$a + bi, a, b \in \mathbb{R} \qquad i^2 = -1$$

$$(a + bi)^* = a - ib$$

$$|a + bi|^2 = (a + bi)(a + bi)^* = a^2 + b^2$$



$$a + bi = re^{i\theta}$$

$$r = |a + bi|$$

$$\tan\theta = \frac{b}{a}$$

Quantum Rule 1

inner product

$$\langle w, v \rangle = \sum_{i=0}^{N-1} w_i^* v_i \qquad v = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \qquad w = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_{N-1} \end{bmatrix}$$

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

"bra"

$$\langle w| = \begin{bmatrix} w_0^* & w_1^* & \cdots & w_{N-1} \end{bmatrix}$$

"ket"

$$\langle w, v \rangle = \langle w|v\rangle = \langle v|w\rangle^* = \sum_{i=0}^{N-1} w_i^* v_i$$

unit vector: $\langle v|v\rangle = 1$

$$\sum_{i=0}^{N-1} v_i^* v_i = \sum_{i=0}^{N-1} |v_i|^2 = 1$$

Example: 2 state device

$$|v\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix}$$

$$\langle v|v\rangle = \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} + \frac{-i}{\sqrt{2}}\frac{i}{\sqrt{2}} = 1$$

Quantum Rule 1

Dirac nota

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} = v_0|0\rangle + v_1|1\rangle + \cdots + v_N|N\rangle$$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad \bullet\bullet\bullet \quad |N-1\rangle = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$U|v\rangle = U\left(v_0|0\rangle + v_1|1\rangle + \cdots + v_N|N\rangle\right)$$

$$U|v\rangle = v_0U|0\rangle + v_1U|1\rangle + \cdots + v_NU|N\rangle$$

“Mathematicians tend to despise Dirac notation, because it can prevent them from making important distinctions, but physicists love it, because they are always forgetting such distinctions exist and the notation liberates them from having to remember.” - David Mermin

Quantum Rule 1, Probabilities?

If we measure our quantum information processing machine, (in the state basis) when our description is $|v\rangle$, then the probability of observing state $|i\rangle$ is $|v_i|^2$.

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \Rightarrow \begin{bmatrix} |v_0|^2 \\ |v_1|^2 \\ \vdots \\ |v_{N-1}|^2 \end{bmatrix} \begin{matrix} Pr(|0\rangle) \\ Pr(|1\rangle) \\ \\ Pr(|N-1\rangle) \end{matrix}$$

requirement of unit vector insures these are probabilities

Example:

$$|v\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$$

$$Pr(|0\rangle) = \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} = \frac{1}{2}$$

$$Pr(|1\rangle) = \frac{-i}{\sqrt{2}}\frac{i}{\sqrt{2}} = \frac{1}{2}$$

Quantum Rule 1, Philosophy

Unfortunately, we often call the unit complex vector, the state of The system. This is like calling the probability distribution the State of the system and confuses our description of the system with the physical state of the system.

For our classical machine, the system is always in one of the states. For the quantum system, this type of statement is much trickier. The only time we will say the quantum system is in a particular state is immediately after we make a measurement of the system.

“I have this student. he's thinking about the foundations of quantum mechanics. He is doomed.“

— John McCarthy (of A.I. fame)

Quantum Rule 1, Nomenclature

Complex unit vector
Vector of amplitudes
Wave function
Quantum State
State

$$|\psi\rangle$$

Actually all of the

$$e^{i\theta}|\psi\rangle$$

are the same description (global phase)

More general condition is wave function is an element of a complex Hilbert space: a vector space with an inner product. We will deal in this class almost exclusively with finite dimensional Hilbert spaces:

$$\mathcal{H} = \mathbb{C}^N$$

Hilbert space
“State space”

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

Rule 2 (Evolution) N x N unitary matrix

The evolution in time of our description of the device is specified by an N x N unitary matrix U, such that if the description of the state $|\psi\rangle$ before the evolution is given by the wave function then the description of the system after this evolution is given by the wave function $U|\psi\rangle$

Quantum Rule 2

before evolution

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle + \cdots + a_{N-1}|N-1\rangle$$

after evolution

$$|\psi'\rangle = b_0|0\rangle + b_1|1\rangle + \cdots + b_{N-1}|N-1\rangle$$

Unitary evolution:

$$|\psi'\rangle = U|\psi\rangle$$

$$\begin{bmatrix} b_0 \\ \vdots \\ b_{N-1} \end{bmatrix} = \begin{bmatrix} U_{0,0} & \cdots & U_{0,N-1} \\ \vdots & & \vdots \\ U_{N-1,0} & \cdots & U_{N-1,N-1} \end{bmatrix} \begin{bmatrix} a_0 \\ \vdots \\ a_{N-1} \end{bmatrix}$$



Unitary matrix

Unitary Matrix?

Unitary N x N matrix: an invertible N x N complex matrix whose inverse is equal to it's conjugate transpose.

Invertible: there exists an inverse of U, such that

$$U^{-1}U = I$$

N x N identity matrix

$$\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & \\ 0 & \cdots & & 1 \end{bmatrix}$$

$$\begin{bmatrix} U_{0,0} & \cdots & U_{0,N-1} \\ \vdots & & \vdots \\ U_{N-1,0} & \cdots & U_{N-1,N-1} \end{bmatrix}^{\dagger} = \begin{bmatrix} U^*_{0,0} & \cdots & U^*_{N-1,0} \\ \vdots & & \vdots \\ U^*_{0,N-1} & \cdots & U^*_{N-1,N-1} \end{bmatrix}$$

$$U^{-1} = U^{\dagger} \quad \text{or} \quad U^{\dagger}U = I$$

Quantum Rule 2, Example

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix}$$

Conjugate:

$$U^* = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$$

Conjugate transpose:

$$U^\dagger = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$$

Unitary?

$$UU^\dagger = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$|\psi\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

evolves to

$$|\psi'\rangle = U|v\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Properties of Unitary Matrices

$$\begin{bmatrix} U_{0,0} & \cdots & U_{0,N-1} \\ \vdots & & \vdots \\ U_{N-1,0} & \cdots & U_{N-1,N-1} \end{bmatrix} \qquad U^\dagger U = I$$

row vectors

$$|r_0\rangle = U_{0,0}|0\rangle + U_{0,1}|1\rangle + \cdots + U_{0,N-1}|N-1\rangle$$

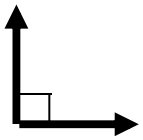
$$|r_1\rangle = U_{1,0}|0\rangle + U_{1,1}|1\rangle + \cdots + U_{1,N-1}|N-1\rangle$$

$$\vdots$$

$$|r_{N-1}\rangle = U_{N-1,0}|0\rangle + U_{N-1,1}|1\rangle + \cdots + U_{N-1,N-1}|N-1\rangle$$

are orthonorm

$$\langle r_i | r_j \rangle = \delta_{i,j} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ otherwise} \end{cases}$$



column vectors are also orthonormal

Special Unitary Matrices

We will often restrict the class of unitary matrices

$$\begin{bmatrix} U_{0,0} & \cdots & U_{0,N-1} \\ \vdots & & \vdots \\ U_{N-1,0} & \cdots & U_{N-1,N-1} \end{bmatrix}$$

to special unitary matrices:

$$\det U = 1$$

U(N) := N x N unitary matrices

SU(N) := N x N special unitary matrices

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

Rule 2 (Evolution) N x N unitary matrix

Rule 3 (Measurement) k measurement operators

Measurements with k outcomes are described by k N x N matrices, M_α which satisfy the completeness criteria:

$$\sum_{\alpha=0}^{k-1} M_\alpha^\dagger M_\alpha = I$$

The probability of observing outcome α if the wave function of the system is $|\psi\rangle$ is given by

$$Pr(\alpha) = \langle\psi|M_\alpha^\dagger M_\alpha|\psi\rangle$$

The new wave function of the system after the measurement is

$$|\psi'\rangle = \frac{M_\alpha|\psi\rangle}{\sqrt{Pr(\alpha)}}$$

Quantum Rule 3

completenes

$$\sum_{\alpha=0}^{k-1} M_\alpha^\dagger M_\alpha = I$$

probability

$$Pr(\alpha) = \langle\psi|M_\alpha^\dagger M_\alpha|\psi\rangle$$

collaps

$$|\psi'\rangle = \frac{M_\alpha|\psi\rangle}{\sqrt{Pr(\alpha)}}$$

probabilities sum to 1:

$$\sum_{\alpha=0}^{k-1} Pr(\alpha) = \sum_{\alpha=0}^{k-1} \langle\psi|M_\alpha^\dagger M_\alpha|\psi\rangle = \langle\psi| \sum_{\alpha=0}^{k-1} M_\alpha^\dagger M_\alpha|\psi\rangle$$
$$= \langle\psi|I|\psi\rangle = 1$$

final state is properly normalized:

$$\langle\psi'| = \frac{\langle\psi|M_\alpha^\dagger}{\sqrt{Pr(\alpha)}} \qquad \langle\psi'|\psi'\rangle = \frac{\langle\psi|M_\alpha^\dagger M_\alpha|\psi\rangle}{Pr(\alpha)} = 1$$

The Computational Basis

We have already described measurements with outcomes

$$|0\rangle, |1\rangle, \ldots, |N-1\rangle$$

Measurement operators:

$$M_0 = |0\rangle\langle 0|, M_1 = |1\rangle\langle 1|, \ldots, M_{N-1} = |N-1\rangle\langle N-1|$$

$$M_\alpha^\dagger M_\alpha = M_\alpha$$

$$M_0 = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & \cdots & 0 \end{bmatrix}$$

Wavefunction $|\psi\rangle$, probability of outcome:

$$Pr(\alpha) = \langle\psi|M_\alpha^\dagger M_\alpha|\psi\rangle = \langle\psi|M_\alpha|\psi\rangle = \langle\psi|\alpha\rangle\langle\alpha|\psi\rangle$$

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle + \cdots + a_{N-1}|N-1\rangle$$

$$Pr(\alpha) = a_\alpha^* a_\alpha = |a_\alpha|^2$$

state of system after measurement is

$$|\psi'\rangle = \frac{|\alpha\rangle\langle\alpha|\psi\rangle}{\sqrt{Pr(\alpha)}} = e^{i\phi}|\alpha\rangle$$

Quantum Rule 3 Example

Measurement operators:

$$M_0 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad M_1 = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Projectors:

$$P^2 = P$$

$$M_0^\dagger M_0 = M_0 \qquad M_1^\dagger M_1 = M_1$$

Completeness:

$$M_0^\dagger M_0 + M_1^\dagger M_1 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Initial state $|\psi\rangle = |0\rangle$

$$Pr(0) = \langle\psi|M_0^\dagger M_0|\psi\rangle = \langle 0|M_0|0\rangle = \frac{1}{2}$$

$$Pr(1) = \langle\psi|M_1^\dagger M_1|\psi\rangle = \langle 0|M_1|0\rangle = \frac{1}{2}$$

Quantum Rule 3 Example

Measurement operators:

$$M_0 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad M_1 = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Initial state $|\psi\rangle = |0\rangle$

outcome 0:

$$|\psi'\rangle = \frac{M_0|\psi\rangle}{\sqrt{Pr(0)}} = \frac{\frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle}{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) =: |+\rangle$$

outcome 1:

$$|\psi'\rangle = \frac{M_1|\psi\rangle}{\sqrt{Pr(1)}} = \frac{\frac{1}{2}|0\rangle - \frac{1}{2}|1\rangle}{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) =: |-\rangle$$

$$M_0 = |+\rangle\langle+| \quad M_1 = |-\rangle\langle-|$$

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

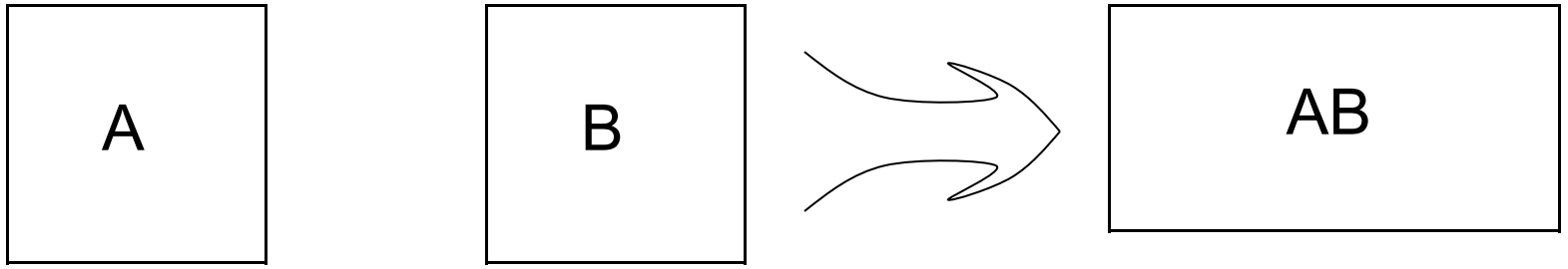
Rule 2 (Evolution) N x N unitary matrix

Rule 3 (Measurement) k measurement operators

Rule 4 (Composite Systems) tensor product

When combining two quantum systems with Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, the joint system is described by a Hilbert space which is a tensor product of these two systems, $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$.

Quantum Rule 4



$\mathcal{H}_A$ $\mathcal{H}_B$ $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$

$|0\rangle$ $|0\rangle$ $|0\rangle \otimes |0\rangle = |00\rangle$

$|1\rangle$ $|1\rangle$ $|0\rangle \otimes |1\rangle = |01\rangle$

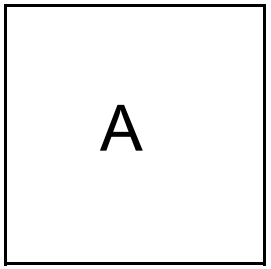
$|1\rangle \otimes |0\rangle = |10\rangle$

$|N\rangle \otimes |M\rangle = |NM\rangle$

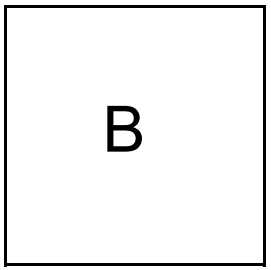
$|N\rangle$ $|M\rangle$

$dim(\mathcal{H}_A) = N$ $dim(\mathcal{H}_B) = M$ $dim(\mathcal{H}_{AB}) = NM$

Quantum Rule 4



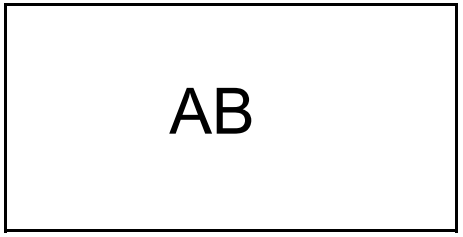
Example :



$$|\psi_A\rangle = \alpha_A|0\rangle + \beta_A|1\rangle$$

$$|\psi_B\rangle = \alpha_B|0\rangle + \beta_B|1\rangle$$





$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle = (\alpha_A|0\rangle + \beta_A|1\rangle) \otimes (\alpha_B|0\rangle + \beta_B|1\rangle)$$

$$= \alpha_A\alpha_B|00\rangle + \alpha_A\beta_B|01\rangle + \beta_A\alpha_A|10\rangle + \beta_A\beta_B|11\rangle$$

separable state

Entangle States

Some joint states of two systems cannot be expressed as

$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$$

Such states are called *entangled states*

Example : $|\psi_{AB}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

We encountered something similar for our probabilistic device:

$$r = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} .$$

Entangled states are, similarly correlated.

But, we will find out later that they are correlated in a very peculiar manner!

Quantum Information Processing Device

Rule 1 (State Description) N states, vector of amplitudes

Rule 2 (Evolution) N x N unitary matrix

Rule 3 (Measurement) k measurement operators

Rule 4 (Composite Systems) tensor product

The Basic Postulates of Quantum Theory

Qubits

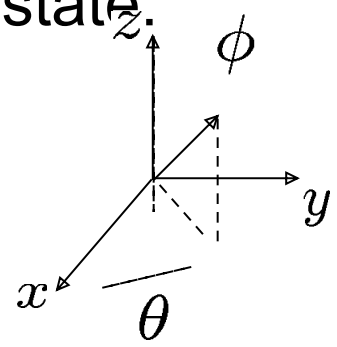


Two level quantum systems

Basis $|0\rangle, |1\rangle$

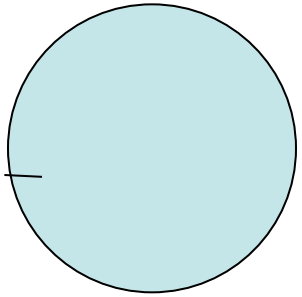
:

Generic state: $|\psi\rangle = \cos\left(\frac{\phi}{2}\right)|0\rangle + e^{i\theta}\sin\left(\frac{\phi}{2}\right)|1\rangle$



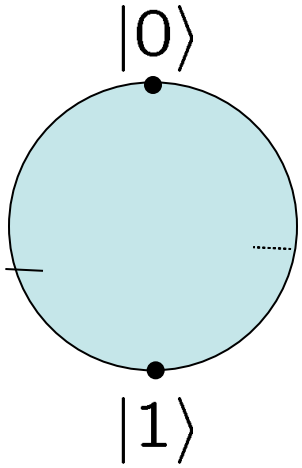
$0 \le \theta < 2\pi$

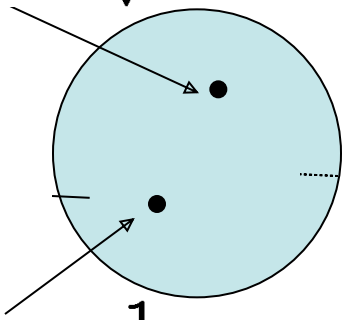
$0 \le \phi \le \pi$

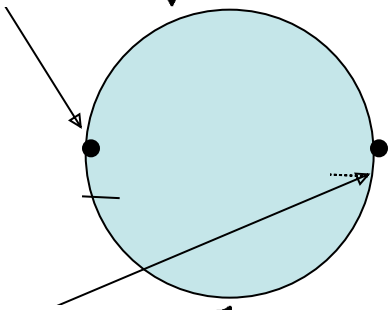


Bloch sphere

$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$







$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$

Pauli Matrices

Important qubit matrices, the Pauli matrices:

$$\sigma_0 = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \sigma_1 = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\sigma_2 = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \qquad \sigma_3 = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Unitary matrices

$$U(n,\theta) = \cos\left(\frac{\theta}{2}\right) I + i \sin\left(\frac{\theta}{2}\right) n \cdot \sigma$$

$0 \leq \theta < 2\pi \qquad n = (n_x, n_y, n_z)$ real unit vector

$$n \cdot \sigma = n_x \sigma_1 + n_y \sigma_2 + n_z \sigma_3$$

Operations on Qubits

$$U(n,\theta) = \cos(\theta/2)I + i\sin(\theta/2)n\cdot\sigma$$

Example:

$$n = (0,0,1) \quad U = \cos(\theta/2)I + i\sin(\theta/2)\sigma_3$$

$$= \begin{bmatrix} \cos(\theta/2) + i\sin(\theta/2) & 0 \\ 0 & \cos(\theta/2) - i\sin(\theta/2) \end{bmatrix}$$

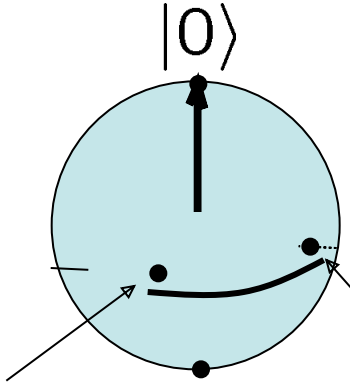
$$= \begin{bmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{bmatrix}$$

$$U|0\rangle = e^{i\theta/2}|0\rangle \sim |0\rangle \qquad U|1\rangle = e^{-i\theta/2}|1\rangle \sim |1\rangle$$

$$U|+\rangle = U\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle) = \frac{1}{\sqrt{2}}\left(e^{i\theta/2}|0\rangle + e^{-i\theta/2}|1\rangle\right)$$

$$\sim \frac{1}{\sqrt{2}}\left(|0\rangle + e^{-i\theta}|1\rangle\right)$$

$$U = \cos(\theta/2) I + i \sin(\theta/2) \sigma_3$$



$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad U|+\rangle = \sim \frac{1}{\sqrt{2}}\left(|0\rangle + e^{-i\theta}|1\rangle\right)$$

U rotates the Bloch sphere about the z-axis

Single qubit rotations:

$$U(n, \theta) = \cos\left(\frac{\theta}{2}\right) I + i \sin\left(\frac{\theta}{2}\right) n \cdot \sigma$$

Rotates by angle θ about the n axis

Some Important Single Qubit Rotations

Hadamard rotation:

Rotation by angle $\frac{\pi}{2}$ about y-axis

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H^2 = I, HXH = Z, HYH = Y$$

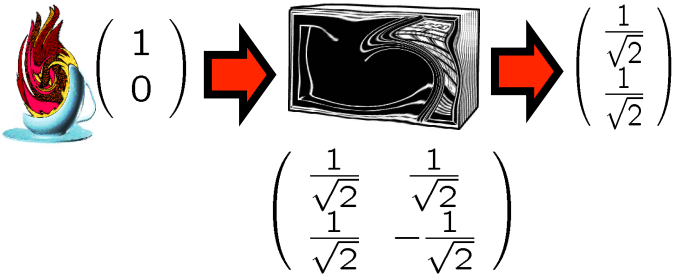
P – gate (also called T – gate):

Rotation by angle $-\frac{\pi}{2}$ about z-axis

$$P = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \sim \begin{bmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

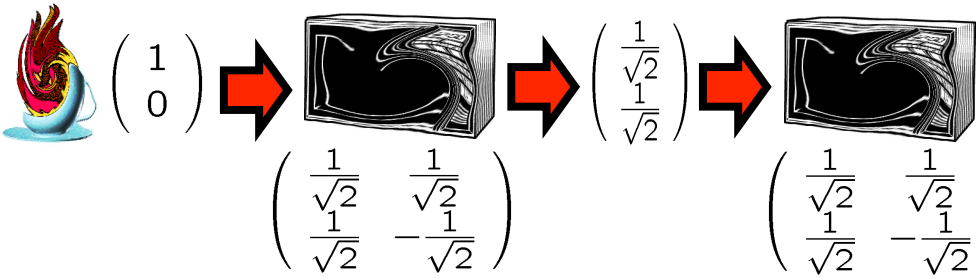
Interference





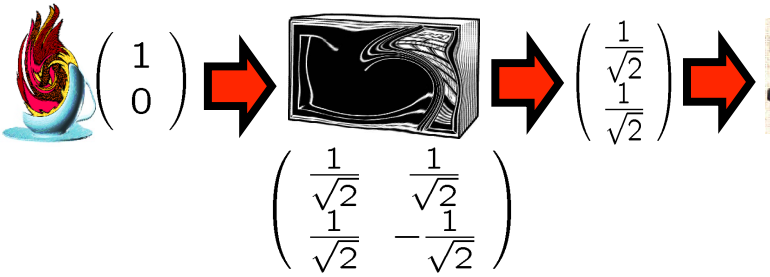
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

50 % **H**
50 % **C**



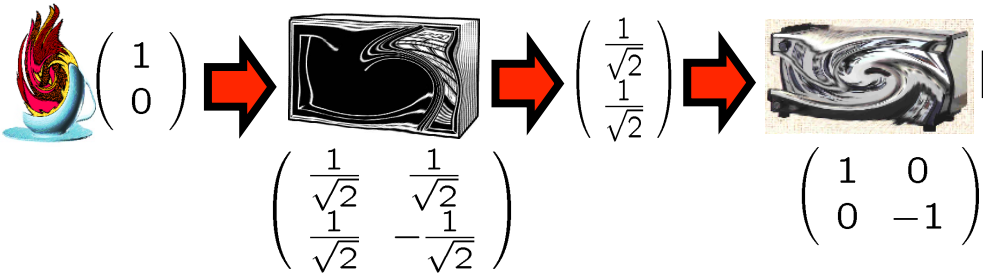
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

100 %
H
0 % **C**



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

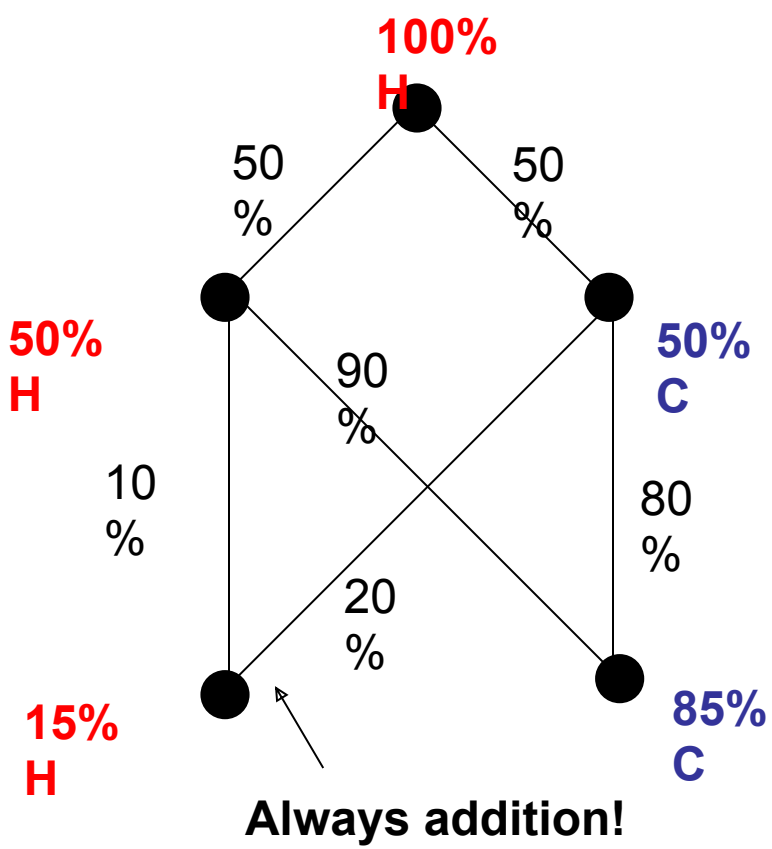
50 % **H**
50 % **C**



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

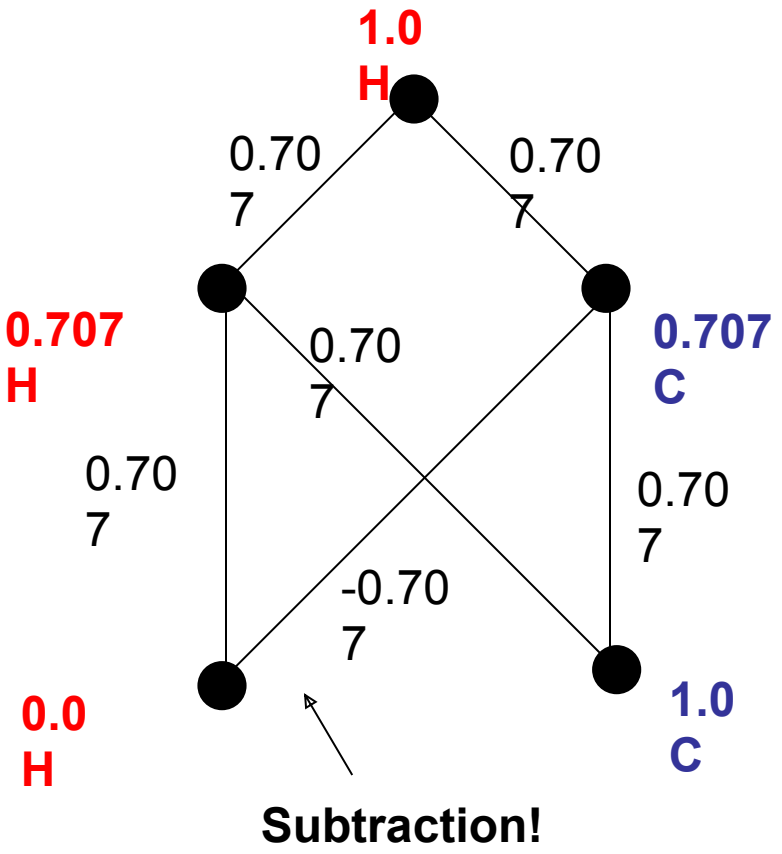
0 % **H**
100 %
C

Interfering Pathways



Always addition!

Classica
l

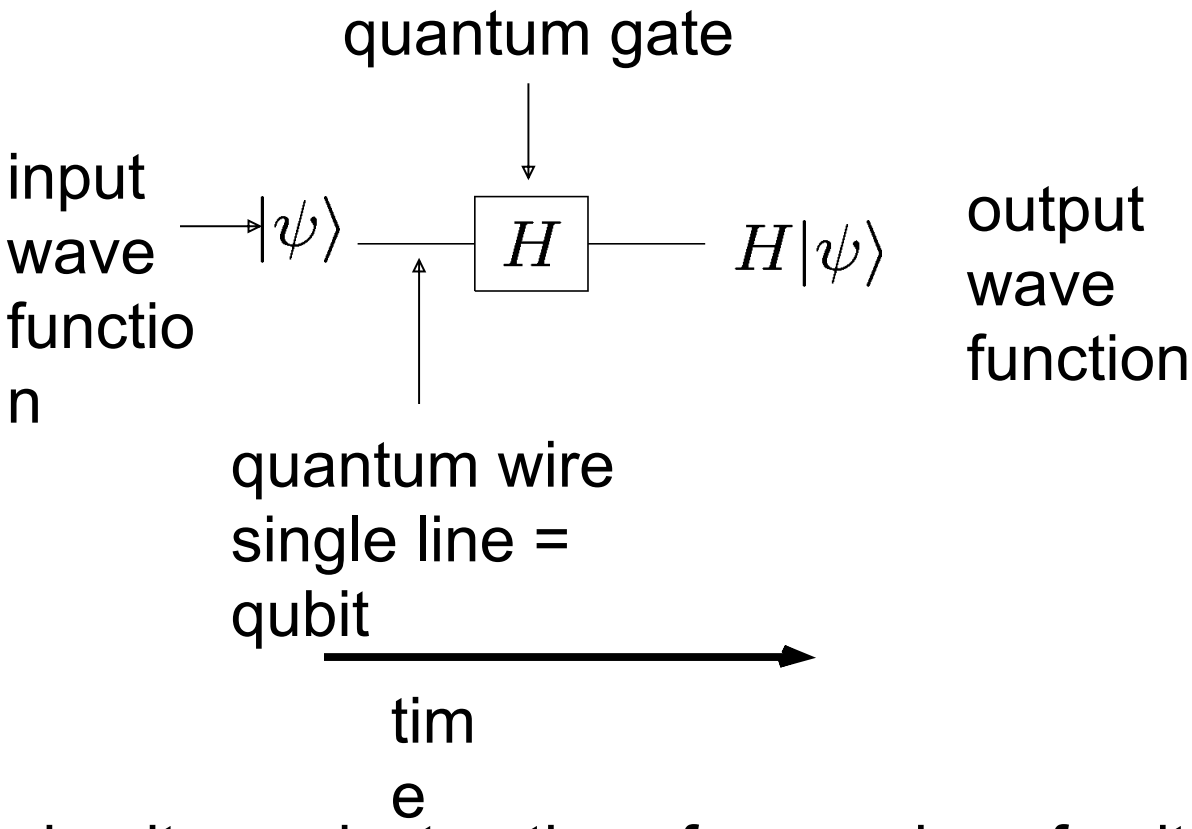


Subtraction!

Quantu
m

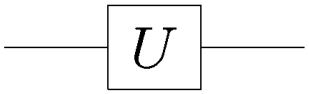
Quantum Circuits

Circuit diagrams for quantum information

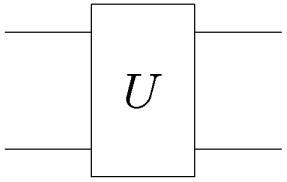


Quantum circuits are instructions for a series of unitary evolutions (quantum gates) to be executed on quantum Information.

Quantum Circuit Elements



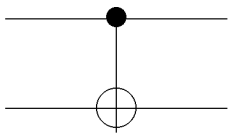
single qubit rotations



two qubit rotations

control

target

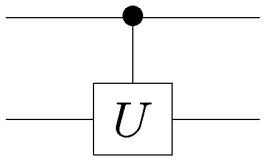


controlled-NOT $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

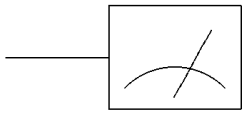
$|00\rangle \rightarrow |00\rangle$
$|01\rangle \rightarrow |01\rangle$
$|10\rangle \rightarrow |11\rangle$
$|11\rangle \rightarrow |10\rangle$

control

target



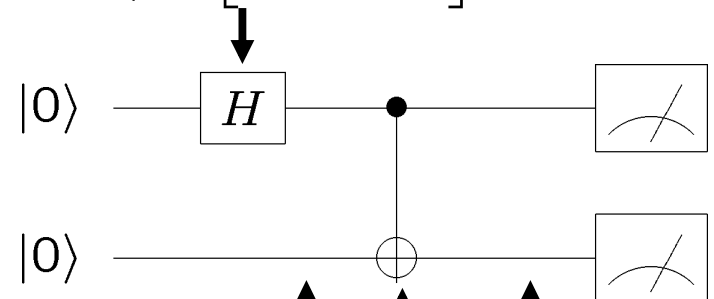
controlled-U $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix}$



measurement in the $|0\rangle, |1\rangle$ basis

Quantum Circuit Example

$$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes I$$



50% $|00\rangle$

50% $|11\rangle$

$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$

$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$|00\rangle \rightarrow |00\rangle$

$|01\rangle \rightarrow |01\rangle$

$|10\rangle \rightarrow |11\rangle$

$|11\rangle \rightarrow |10\rangle$

Deutsch's Problem

A one bit function:

$$f : \{0, 1\} \rightarrow \{0, 1\}$$

Four such functions:

$f(0) = f(1) = 0$ "constant"

$f(0) = f(1) = 1$

$f(0) = 0, f(1) = 1$ "balanced"

$f(0) = 1, f(1) = 0$

Deutsch's Problem

instance: unknown function f

problem: determine whether function is constant or balanced

Classical Deutsch's Problem

$$f(0) = f(1) = 0$$ "constant"

$$f(0) = f(1) = 1$$

$$f(0) = 0, f(1) = 1$$ "balanced"

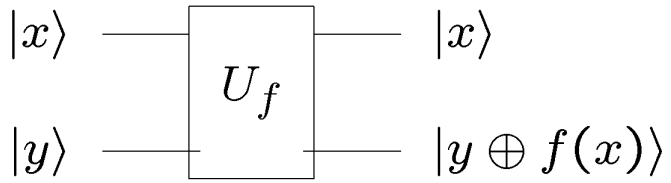
$$f(0) = 1, f(1) = 0$$

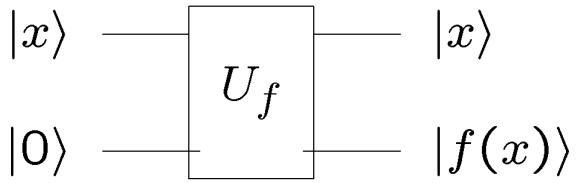
Question: What is $f(0)$?

Must ask two question to separate balanced from constant.

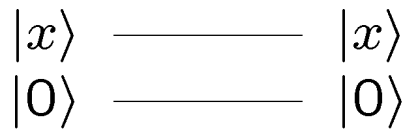
Deutsch's Problem

Oracle :

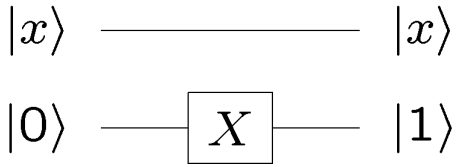




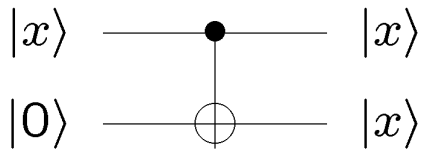
$f(0) = f(1) = 0$



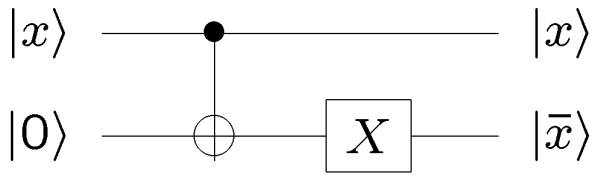
$f(0) = f(1) = 1$



$f(0) = 0, f(1) = 1$



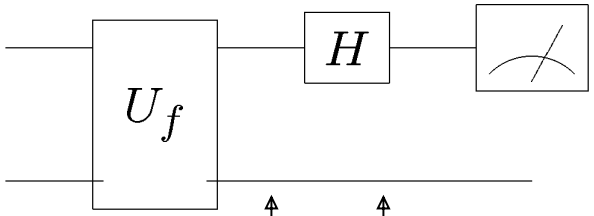
$f(0) = 1, f(1) = 0$



If the wires and gates are classical, then we need two queries.
What if the wires and gates are quantum?

Quantum Deutsch's Problem

$$\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$



$$\frac{1}{2}(|0f(0)\rangle - |0\bar{f}(0)\rangle + |1f(1)\rangle - |1\bar{f}(1)\rangle)$$

constant $f(0) = f(1)$

$$\frac{1}{2}(|0\rangle + |1\rangle)(|f(0)\rangle - |\bar{f}(0)\rangle) \quad |0\rangle\frac{1}{\sqrt{2}}(|f(0)\rangle - |\bar{f}(0)\rangle)$$

balanced $f(0) = \bar{f}(1)$

$$\frac{1}{2}(|0\rangle - |1\rangle)(|f(0)\rangle - |\bar{f}(0)\rangle) \quad |1\rangle\frac{1}{\sqrt{2}}(|f(0)\rangle - |\bar{f}(0)\rangle)$$

Measure first qubit determines constant vs. balance in 1 query!

THE BEGINNING OF QUANTUM COMPUTING

Linear Algebra

Matrices:

Eigenvectors, eigenvalues

$$M|\psi\rangle = \lambda_\psi|\psi\rangle$$

Characteristic equation

$$\det(M - \lambda I) = 0$$

solve for eigenvalues

use eigenvalues to determine eigenvectors

Example:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\det(X - \lambda I) = 0$$

$$\det \begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = 0$$

$$\lambda^2 - 1 = 0, \lambda = \pm 1$$

$\lambda = 1$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$a = b$ $\begin{bmatrix} a \\ a \end{bmatrix}$

Linear Algebra

Matrices continued

Hermitian: $H = H^\dagger$

eigenvalues are real $\quad H|\psi\rangle = \lambda_\psi |\psi\rangle \quad \lambda_\psi \in \mathbb{R}$

diagonalizing Hermitian matrix:

$$UHU^\dagger = \mathrm{diag}(\lambda_0, \lambda_1, \ldots, \lambda_{N-1}) = \begin{bmatrix} \lambda_0 & 0 & \cdots & 0 \\ 0 & \lambda_1 & & \vdots \\ \vdots & & \ddots & \\ 0 & \cdots & & \lambda_{N-1} \end{bmatrix}$$

U is unitary $\quad UU^\dagger = U^\dagger U = I$

rows of U are eigenvectors of H

Linear Algebra

Normal Matrices: $NN^\dagger = N^\dagger N$

Spectral Theorem: A matrix is diagonalizable iff it is normal

Implies both unitary and Hermitian matrices are diagonalizable.

Eigenvalues of unitary matrices:

$$UU^\dagger = I$$

in basis where U is diagonal, $U = \mathrm{diag}(\lambda_0, \ldots, \lambda_{N-1})$ this implies

$$\lambda_i \lambda_i^* = 1 \Rightarrow \lambda_i = e^{i\phi_i}$$

Linear

Example :

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\det(X - \lambda I) = 0$$

$$\det \begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = 0$$

$$\lambda^2 - 1 = 0, \lambda = \pm 1$$

$\lambda = 1$ eigenvector: $\begin{bmatrix} a \\ a \end{bmatrix}$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$\lambda = -1$ eigenvector: $\begin{bmatrix} a \\ -a \end{bmatrix}$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$UXU^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Linear Algebra

Trace

Sum of the diagonal elements of a matrix:

$$\mathrm{Tr}[A] = \sum_{i=0}^{N-1} A_{i,i} \qquad \mathrm{Tr}[A+B] = \mathrm{Tr}[A] + \mathrm{Tr}[B]$$

$$\mathrm{Tr}[AB] = \sum_{i,j=0}^{N-1} A_{i,j} B_{j,i} = \mathrm{Tr}[BA]$$

Suppose H is Hermitian

$H = U^\dagger D U$ D is diagonal

$$\mathrm{Tr}[H] = \mathrm{Tr}[U^\dagger D U] = \mathrm{Tr}[U U^\dagger D] = \mathrm{Tr}[D] = \sum_{i=0}^{N-1} \lambda_i$$

Trace is the sum of the eigenvalues

Linear Algebra

Determinant

$$\mathrm{Det}[A] = \sum_{\pi} (-1)^{|\pi|} \prod_{j=0}^{N-1} A_{i,\pi(j)}$$

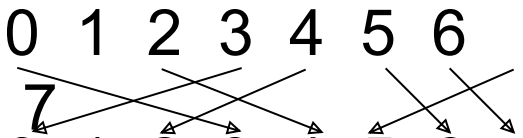
π permutation of 0,1,…,N-1

$|\pi| =$ number of transpositions

$$\mathrm{Det}[AB] = \mathrm{Det}[A]\mathrm{Det}[B]$$

Example

$\pi = (03)(24)(567)$



$|\pi| = 4$

$$\mathrm{Det}[UBU^{-1}] = \mathrm{Det}[U]\mathrm{Det}[B]\mathrm{Det}[U^{-1}]$$
$$= \mathrm{Det}[B]\mathrm{Det}[UU^{-1}] = \mathrm{Det}[B]$$

Suppose H is H

$$\mathrm{Det}[H] = \mathrm{Det}[D] = \prod_{i=0}^{N-1} \lambda_i$$ product of eigenvalues

Linear Algebra

Singular value decomposition:

not all matrices has full set of eigenvectors

Example:

$$M = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \qquad M|1\rangle = 0|1\rangle$$

but every matrix has a singular value decomposition

$$M = UDV \qquad UU^\dagger = VV^\dagger = I$$

D diagonal

Example:

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Linear Algebra

Matrix exponentiation:

$$\exp[M] = I + M + \frac{M^2}{2} + \frac{M^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{M^k}{k!}$$

if $M = U^{-1} D U \quad D = \mathrm{diag}(\lambda_0, \ldots, \lambda_{N-1})$

$$\exp[M] = I + U^{-1} D U + \frac{(U^{-1} D U)^2}{2} + \frac{(U^{-1} D U)^3}{3!} + \cdots$$

$$= U^{-1} \left[I + D + \frac{D^2}{2} + \frac{D^3}{3!} + \cdots \right] U$$

$$= U^{-1} \left[\mathrm{diag}(e^{\lambda_0}, \ldots, e^{\lambda_{N-1}}) \right] U$$

Linear Algebra

Example:

$$\exp[i\theta X] \qquad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$H^\dagger (i\theta X) H = \begin{bmatrix} i\theta & 0 \\ 0 & -i\theta \end{bmatrix} \qquad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\exp[i\theta X] = H^\dagger \exp\left[\begin{bmatrix} i\theta & 0 \\ 0 & -i\theta \end{bmatrix}\right] H^\dagger$$

$$= H^\dagger \exp \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} H^\dagger$$

$$= \frac{1}{2} \begin{bmatrix} e^{i\theta} + e^{-i\theta} & e^{i\theta} - e^{-i\theta} \\ e^{i\theta} - e^{-i\theta} & e^{i\theta} + e^{-i\theta} \end{bmatrix} = \begin{bmatrix} \cos(\theta) & i\sin(\theta) \\ i\sin(\theta) & \cos(\theta) \end{bmatrix}$$

Linear Algebra

Special case of

$$\exp[i\theta M] = I + i\theta M + \frac{(i\theta M)^2}{2} + \frac{(i\theta M)^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{(i\theta M)^k}{k!}$$

when $M^2 = I$

$$\exp[i\theta M] = I + i\theta M + \frac{-\theta^2 I}{2} + \frac{-i\theta^3 M}{3!} + \cdots$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{(2k)!} I + i \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k+1}}{(2k)!} M$$

$$= \cos(\theta) I + i \sin(\theta) M$$

Hamiltonian

Rule 2 (Evolution) N x N unitary matrix

The evolution in time of our description of the device is specified by an N x N unitary matrix U, such that if the description of the state $|\psi\rangle$ before the evolution is given by the wave function then the description of the system after this evolution is given by the wave function $U|\psi\rangle$

Rule 2 prime: (Hamiltonian Evolution)

The evolution of our description of the device in time is specified by a possibly time dependent N x N matrix known as a Hamiltonian. If the wave function is initially $|\psi\rangle$ then after a time t the new state is $U|\psi\rangle$ where

$$U = T \exp\left(-i \int_0^t H(t) dt\right)$$

Hamiltonians

Where we hide the physics:

$$U = T \exp\left(-i \int_0^t H(t) dt\right)$$

time ordering

Time independent Hamiltonian:

$$U = \exp\left(-iHt\right)$$

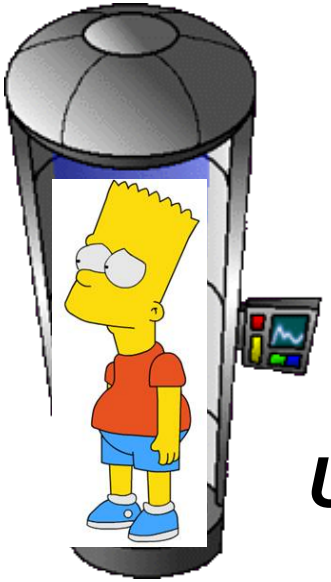
Eigenstates of Hamiltonian are the energy eigenstates.

$$H|i\rangle = E_i|i\rangle$$

$E_i :=$ energies

The Next Episode

Teleportation



Superdense Coding

Universal Quantum Computers



Density Matrices